

Course overview

Modeling the distribution of marine biodiversity under global climate change

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Scope

Provide students **training in the scope of Macroecology and species distribution modeling** to infer and discuss the **interactions** and potential **impacts of global climate changes on marine biodiversity**.

It is mostly **hands-on oriented**, with a strong component on:

- 1) biodiversity and climatic data acquisition, management and visualization (e.g., species occurrence data and the new Shared Socioeconomic Pathway scenarios of climate change);
- 2) species distribution modeling frameworks using state of the art algorithms (e.g., machine learning extreme gradient boosting).

Part 1 : Lectures and hands-on oriented coding in R computing language (22 Apr - 28 Apr).

Part 2 : Written assignment that is an individual research study addressing:

(1) the interactions between, or

(2) the impacts of, global climate changes on marine biodiversity.

Part 3 : Oral plenary presentation based on the individual assignment (8 May - 9 May).

Total workload estimated to approximately 120 hours including assignment preparations.

Sessions (theory)

T1 Ecological niches and geographic distributions, a perspective from macroecology

T2 Climate oscillations and distributional shifts of marine biodiversity

T3 Projected impacts of future climate change on marine biodiversity

T4 Principles of Geographic Information Systems

T5 Principles of species distribution modeling

T6 Potential applications of SDM

T7 Biological and environmental data for SDM

T8 Evaluating predictive performance and setting decision thresholds

T9 Improving transferability of SDM

T10 Climate change ecology informing conservation

T11 FAIR Research

Sessions (practice)

P1 Introduction to R and RStudio

P2 R as a Geographic Information System

P3 Downloading and managing biodiversity data in R

P4 Downloading and managing environmental data in R

P5 Developing Species Distribution Models in R

P6 Performance of Species Distribution Models

P7 Data-driven hyperparameter and monotonic constraints

P8 The diversity of algorithms and ensembles

Assessment methods and criteria

Written assignment

(grading scale: approved/ not approved; to be delivered on the **6 May**):

- . identifying the main environmental drivers shaping species distributions;
- . predicting present-day distribution;
- . projecting future distribution shifts;
- . predicting marine invasion processes, etc.

Oral presentation

(up to 10 minutes; on the **8th May**)

Oral exam focusing the written assignment

(up to 10 minutes; grading scale: approved/ not approved; on the **9th May**).

Resources

<https://github.com/jorgeassis/courseModellingDistributionBiodiversity>

Sessions (PDFs), Data, Code, Challenges, Solutions and Recipes.

Relevant questions

[book a meeting] jorgemfa@gmail.com